density is as a stationary point of the Gibbs-Boltzmann entropy[4]

$$\mathrm{S} = -\mathrm{k_B} \int \rho \ln \rho \mathrm{d}\omega,$$

subject to the constraints

$$\int \rho(z)\mathrm{d}\omega = 1, \qquad \int H(z)\rho(z)\mathrm{d}\omega = \bar{\mathrm{E}},$$

where the first constraint amounts to a normalization to make ρ a probability density, whereas the second fixes the average energy of the system to a predefined value. It does not matter what this value is at this point.

This can be seen as a variational principle. Using the method of Lagrange multipliers, we seek stationary solutions of

$$V = \mathrm{S}/\mathrm{k_B} - \lambda \left[\int \rho(z)\mathrm{d}\omega - 1 \right] - \beta \left[\int H(z)\rho(z)\mathrm{d}\omega - \bar{\mathrm{E}} \right]$$

with respect to the choice of density ρ and coefficients λ and β. The resulting equations for the maximizing density are obtained by variational differentiation (see Sect. 2.2.1):

$$\frac{\delta V}{\delta \rho} = 0$$

$$\int \rho(z)\mathrm{d}\omega = 1$$

$$\int H(z)\rho(z)\mathrm{d}\omega = \bar{\mathrm{E}}.$$

The first equation reduces to

$$-(1 + \ln \rho) - \lambda - \beta H = 0$$

yielding

$$\rho = e^{-1-\lambda} e^{-\beta H}.$$

[4]This form of the entropy is more general than the "Boltzmann" form (6.1); it was introduced in Gibbs' treatise [147].